



Project presentation

Presentation at INVICTA

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Co-funded by
the European Union

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Project Overview

- Consortium
 - 16 partners
 - 10 academic (2 from USA)
 - 6 companies
- Budget: 8M€
- Call: HORIZON-INFRA-TECH
- Feb 2023 - Jan 2026



Partner short name	Type
INESC TEC	Academic
UOULU	Academic
BSC CNS	Academic
EURECOM	Academic
SORBONNE	Academic
INRIA	Academic
CSC	Academic
Queen's University Belfast	Academic
ALLBESMART LDA	Company
GREENERWAVE	Company
Adapttech	Company
Finwe Oy	Company
FinCloud Ltd.	Company
INTERDIGITAL	Company
Rice University	Academic - Associated P.
RUTGERS University	Academic - Associated P.

RI	Location	Partners	
5GTN	Distributed in Finland	UOULU	SLICES-RI
Open5GLab	Sophia Antipolis, France	EURECOM	
FIT-R2lab	Sophia Antipolis, France	INRIA	
MareNostrum 4	Barcelona, Spain	BSC	PRACE-RI
Mahti	Kajaani, Finland	CSC	
CONVERGE-PT	Porto, Portugal	INESC TEC	
CWI mmWave Lab	Belfast, UK	QUB	USA
COSMOS	New York, USA	RUTGERS	
Powder-Renew	Salt Lake City, USA	Rice	

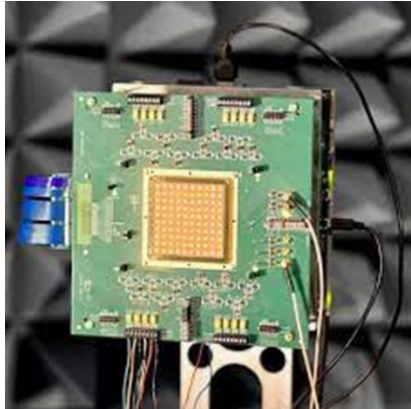
CONVERGE Advisory Board

- José Pedro Borrego (ANACOM, Portugal)
- Andreas Mueller (Bosch/5G-ACIA, Germany)
- Olli Liinamaa (Nokia, Finland)
- Naser Damer (Fraunhofer IGD, Germany)
- Carlos Cordeiro (Intel, USA)
- Sumit Roy (Washington Univ., USA)

Motivation

- **Communications** and **Vision** have evolved as distinct scientific fields

Broadband Internet
Mobile connectivity
Localisation and sensing
Higher frequencies (LoS)



Source: COSMOS Testbed

millimetre-wave radios

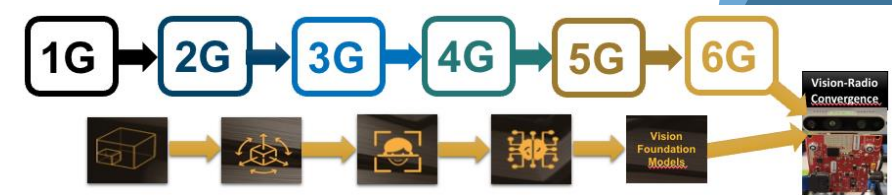


Source: Entropix/Boulder AI

Machine vision/cameras

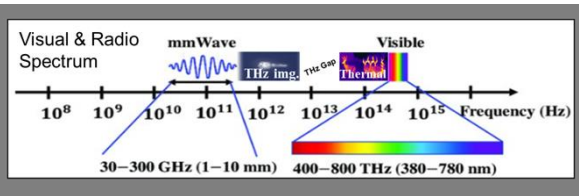
Recognition of forms/text
Facial recognition
Object detection
3D reconstruction
World segmentation
Foundation models

- **Communications** and **vision** are addressing common problems:
localization, sensing, distributed computing, increased intelligence
- These problems should be tackled together



Motivation

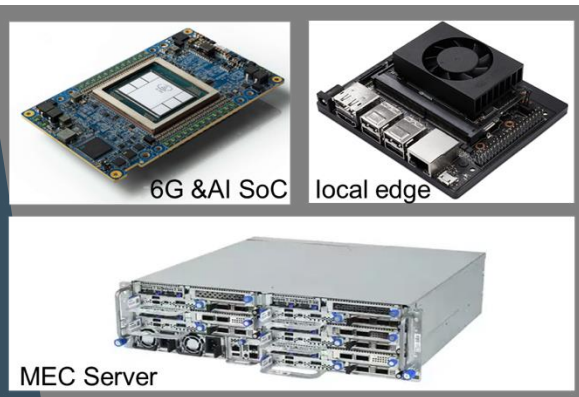
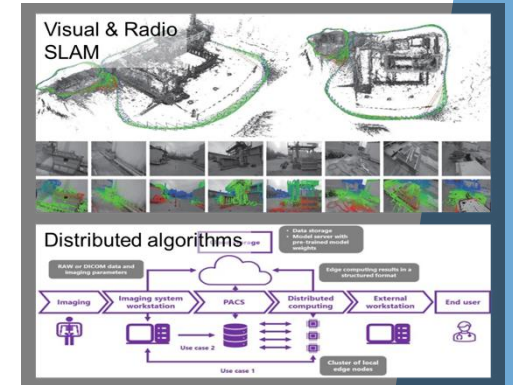
► Convergence between radio and vision



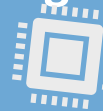
Spectrum level:
5G/6G higher frequencies
Computer vision long wavelengths



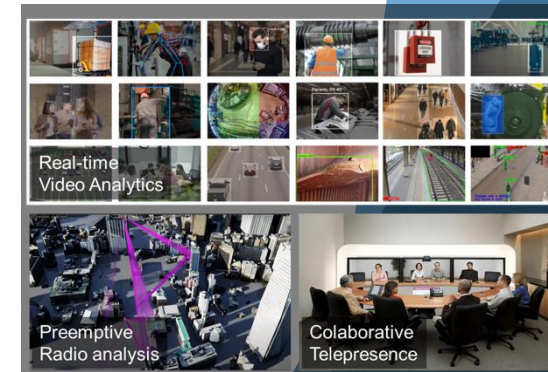
Algorithmic level:
Localization, mapping, reconstruction
Distributed collaborative algorithms



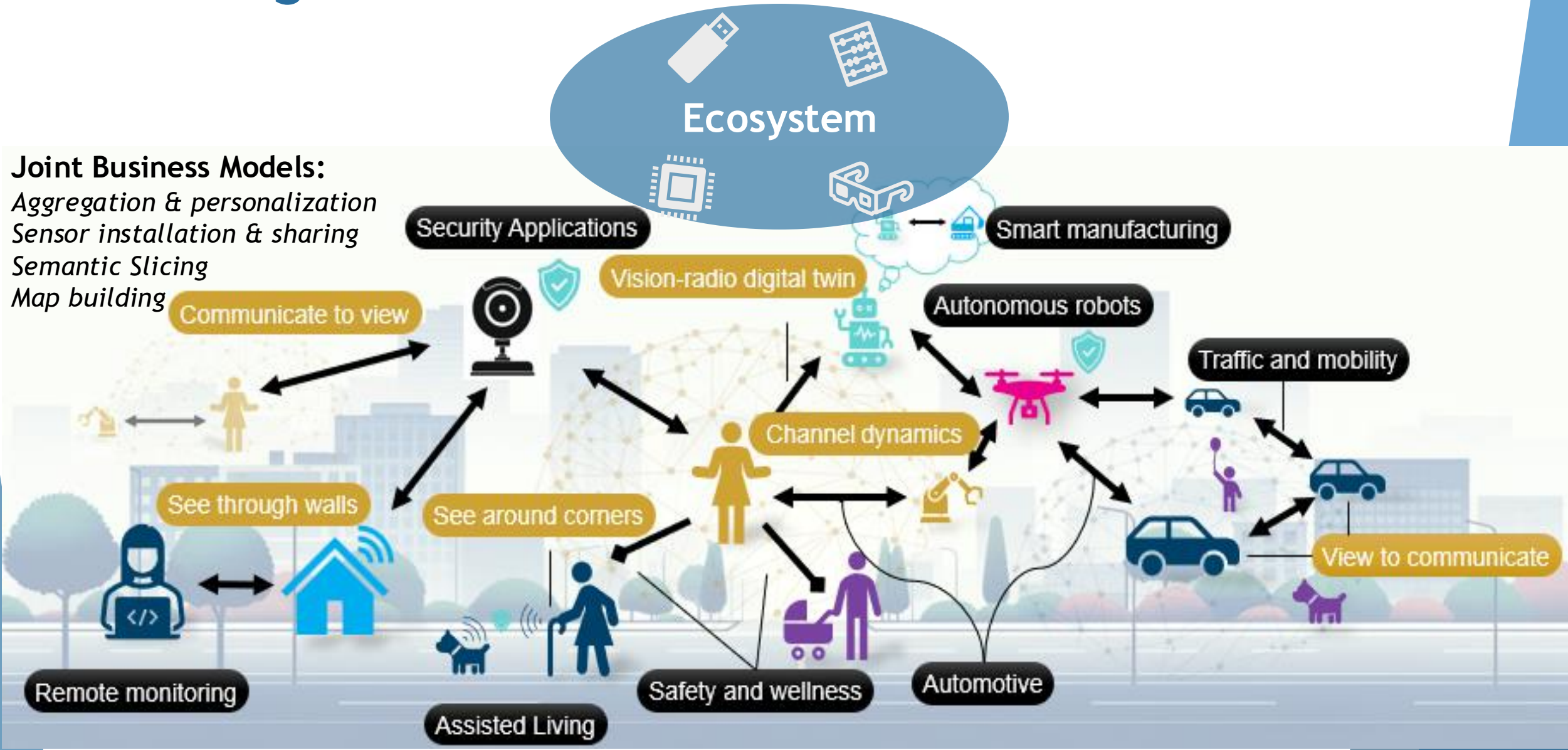
Hardware level:
Multiprocessors (Sensor, AI, DSP)
Edge-to-cloud computing



Application level:
Enabling real time vision
Preemptive radio analysis
AR/VR



Convergence at several levels



CONVERGE objectives

- ▶ Development of an innovative toolset **combining radio and vision based communications and sensing technologies** to enable an emerging area of research
 - “view-to-communicate and communicate-to-view”
- ▶ **Advance the state of the art of Research Infrastructures (RIs) aligned with the ESFRI SLICES-RI**
- ▶ Provide the scientific community with **open datasets** of experimental and simulated data
- ▶ Creating new knowledge and discoveries at the **intersection of**
 - ▶ wireless communications
 - ▶ computer vision
 - ▶ sensing
 - ▶ machine learning

View-to-communicate

Vision-driven wireless dynamics prediction allows us to preemptively anticipate changes in the communications channel by observing the surrounding environment

Interpreting Environment:

Computer vision reconstructs the environment

Dynamic Adjustments:

Anticipating obstructions and adapting network

Predictive Models:

Deterministic estimations obtained by ray-tracing radio signals in a 3D environment model

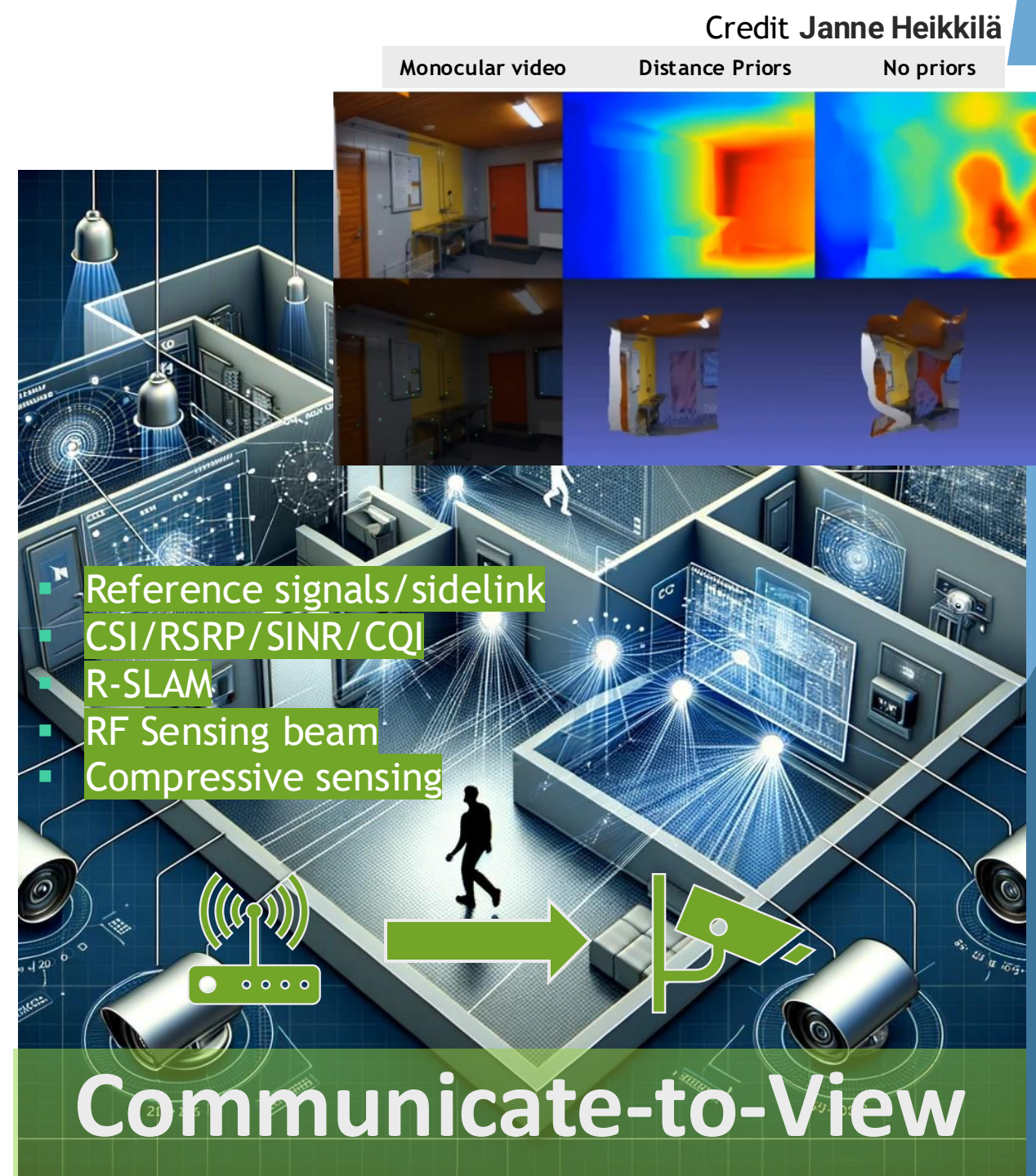
Shifting from reactive to proactive networks towards Dynamic Network Management



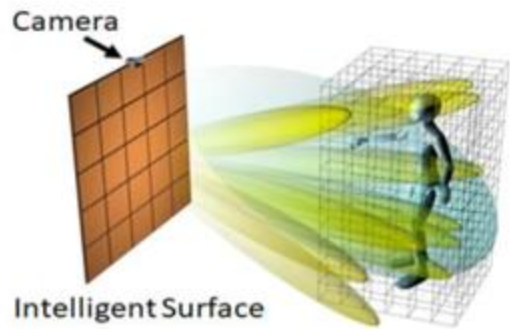
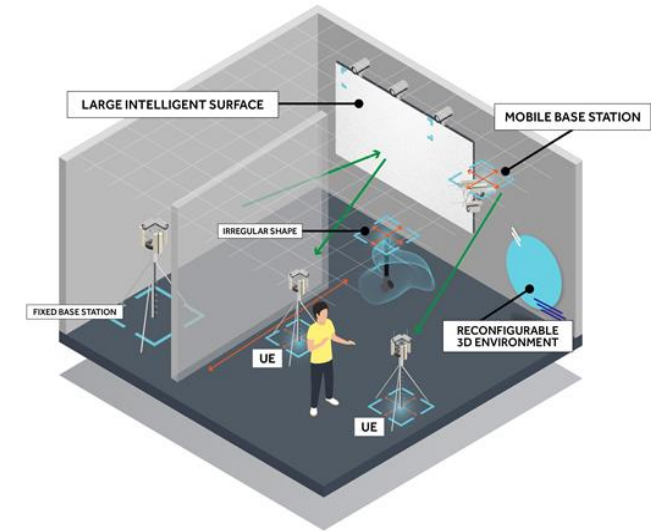
Communicate-to-View

Communication data can enhance situational awareness

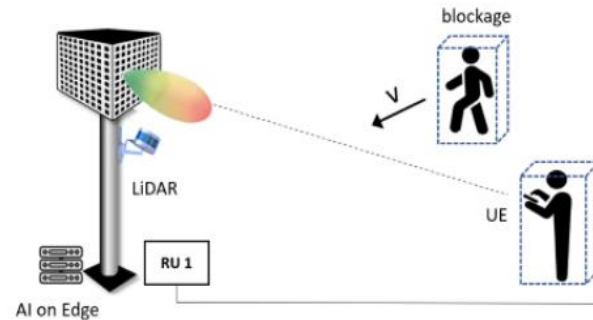
- ▶ Provide approximate distance priors that can be useful in monocular depth estimation
- ▶ Improve depth/distance estimation for novel viewpoints from incomplete data/occluded spaces
- ▶ Extend visual reach beyond physical barriers: through walls, around corners
- ▶ Real-time 3D mapping & visualization techniques enabling applications ranging from augmented reality to autonomous navigation.



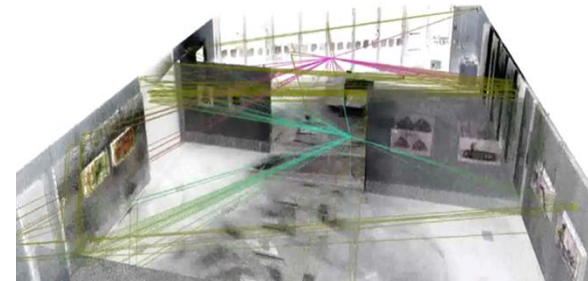
4 Main tools proposed



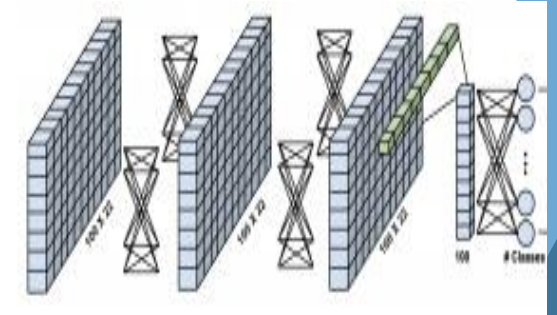
Vision-aided RIS



Vision-aided base station



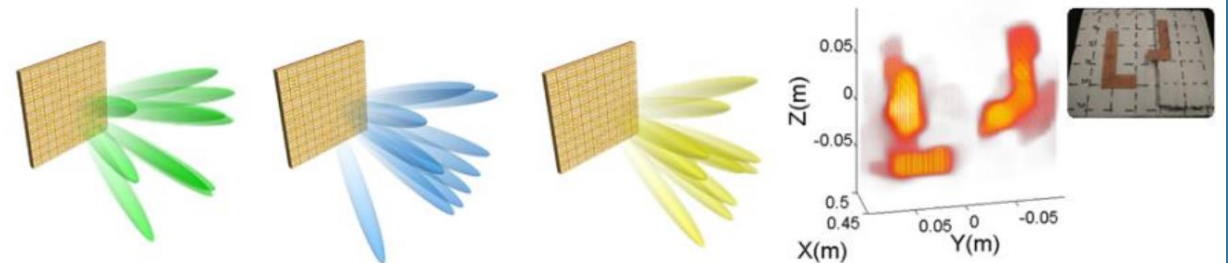
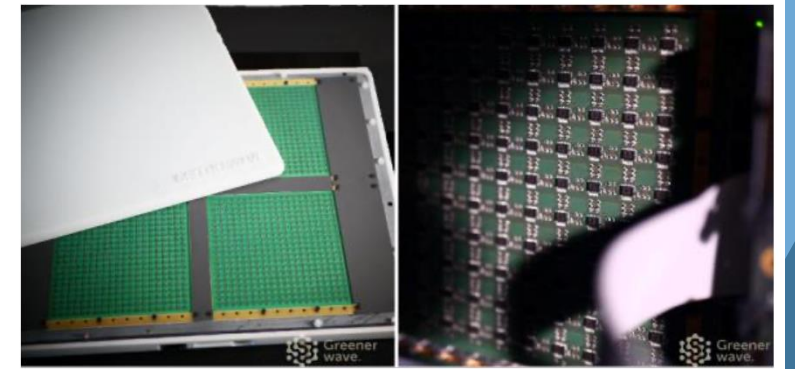
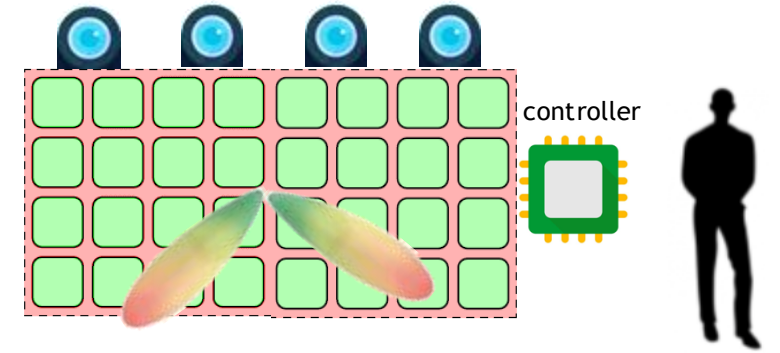
**Vision-radio simulator
(Digital Twin)**



ML for multimodal data

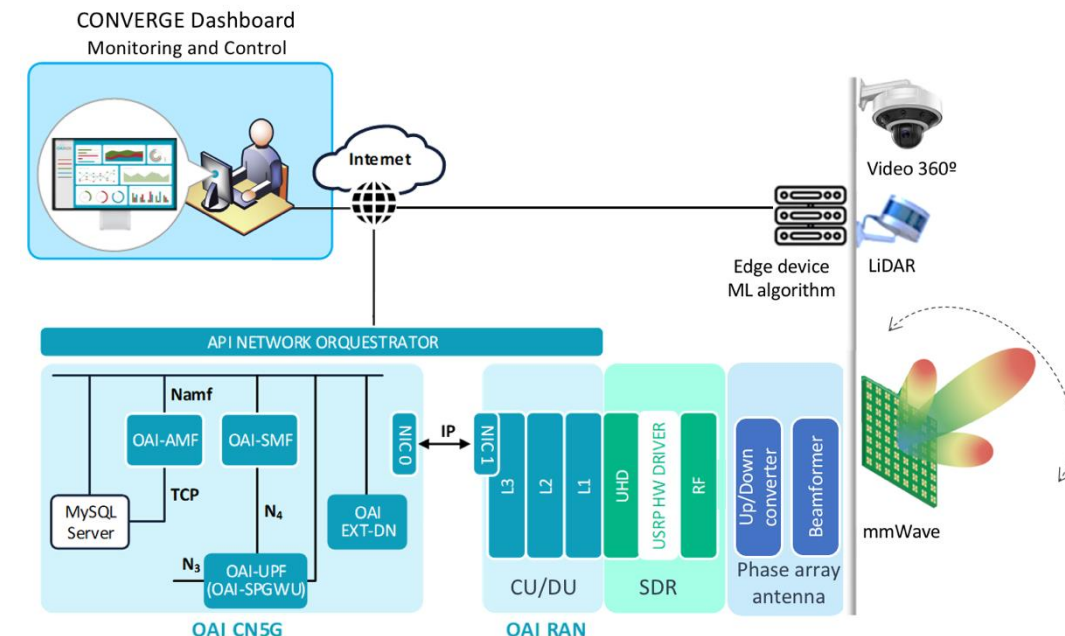
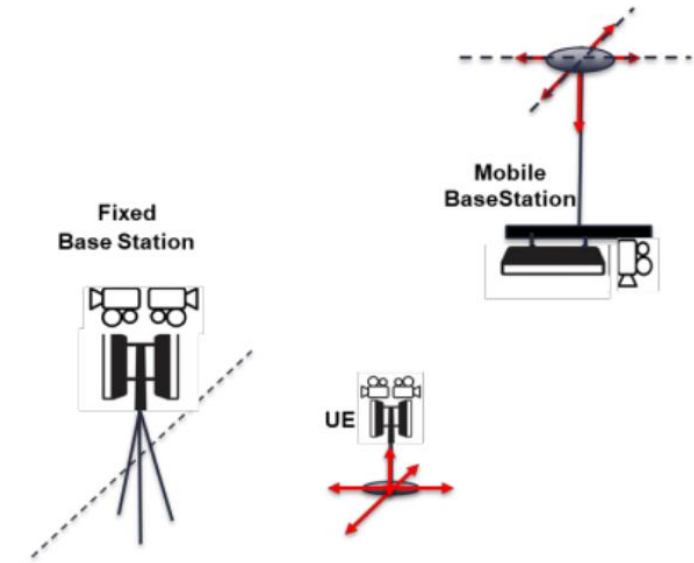
1. Large vision-aided intelligent surface

- ▶ Main purpose: enable the R&D striving towards the **vision of smart radio environments**.
- ▶ Combination of a smart programmable meta-surface antenna with a camera array
- ▶ Experimentation of
 - ▶ MIMO wireless communications
 - ▶ ultra-high precision 3D positioning and environment sensing, including human sensing and holography
- Main requirements:
 - FR1 and FR2, BW: 100 MHz and 400 MHz
 - Expose control of RIS phase profile
 - Communications and sensing specific profiles
 - Multiple camera views



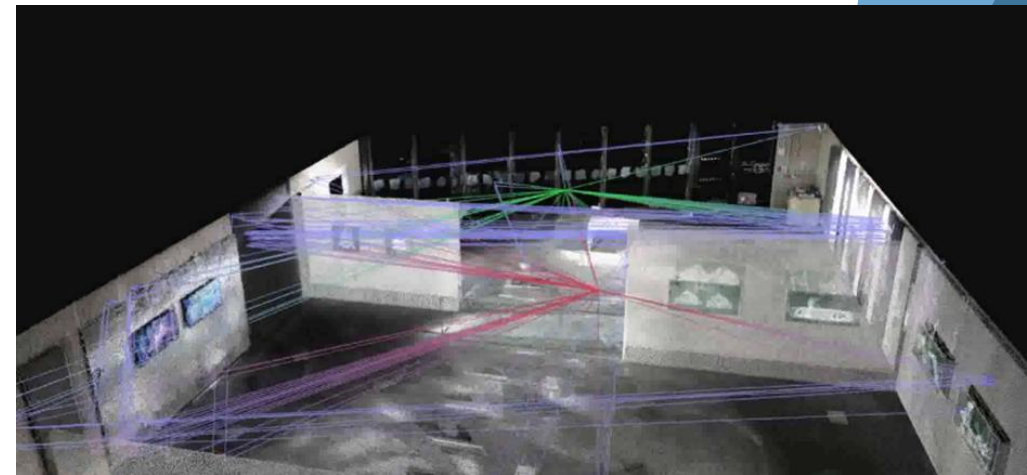
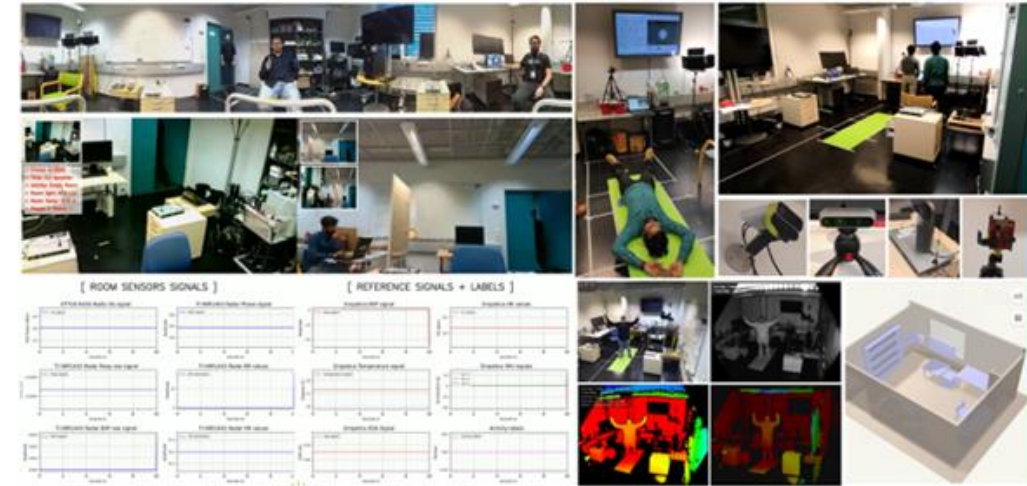
2. Vision-aided base station

- ▶ Enable experimentation with mobile terminals
 - ▶ beamforming
 - ▶ multi-user access
 - ▶ opportunistic scheduling
 - ▶ Radio and vision sensing
- ▶ Fixed and mobile Base station
 - ▶ controlled mobility
- ▶ Mobile user equipment (UE - terminals)
- ▶ Main requirements
 - ▶ Mobile instance based on robotic arm
 - ▶ FR1 and FR2, BW: 100 MHz and 400 MHz
 - ▶ Expose antenna array beam control
 - ▶ O-RAN and Open Air Interface compatible
 - ▶ Communications and radio/vision sensing configurations



3. 3D environment modeller and vision-radio simulator

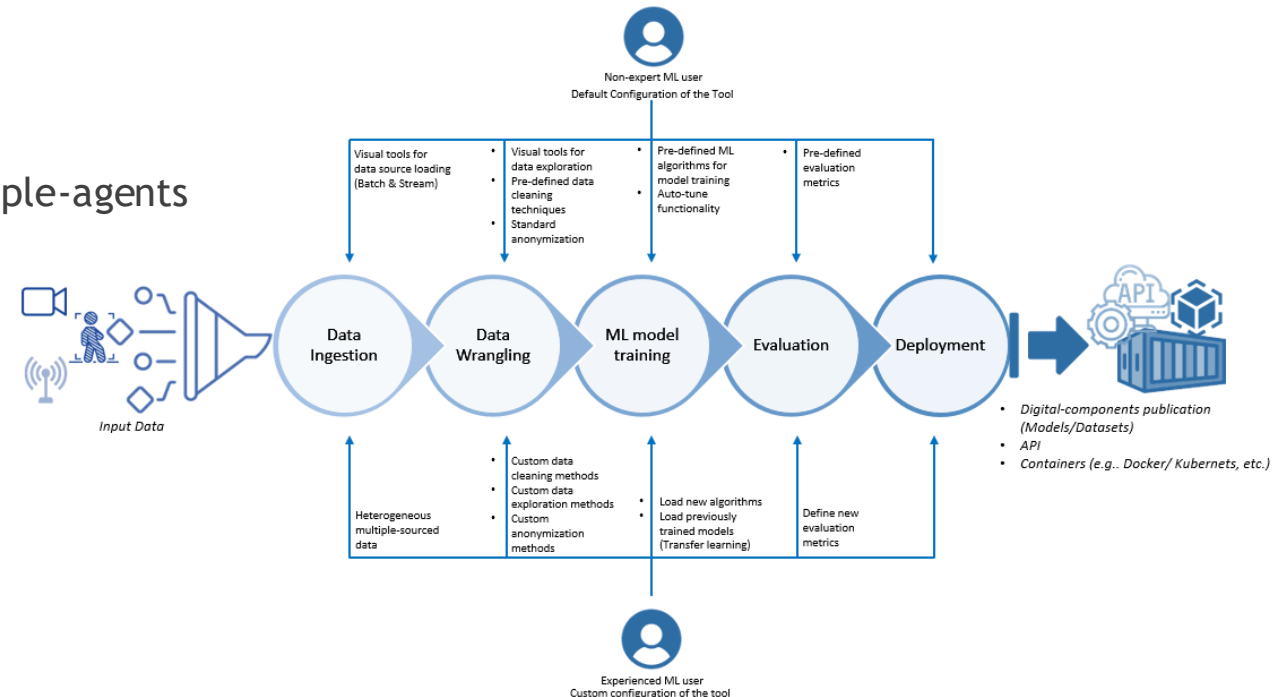
- ▶ **Digital twin of the experimental chamber**
- ▶ **3D environment modeller**
 - ▶ to create a digital 3D representation of the environment
- ▶ **Vision-radio simulator**
 - ▶ to simulate observations at any location, and simulate radio signal propagation
- ▶ Enables a detailed **planning of the experiments**
- ▶ Main requirements
 - ▶ 3D modelling with accurate dimensions and textures
 - ▶ Point clouds and triangle meshes
 - ▶ Account interaction of radio waves with materials
 - ▶ Integration of data from wireless sensors
 - ▶ Simulate different object and furniture placements and layout changes



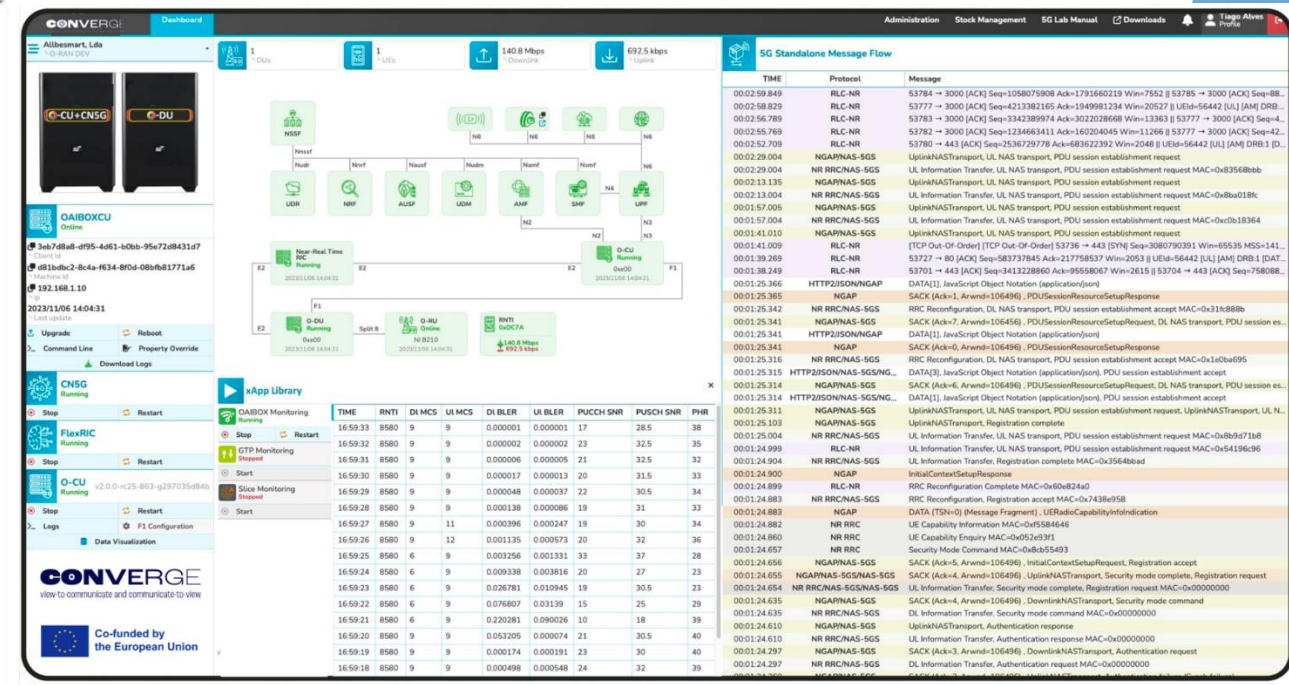
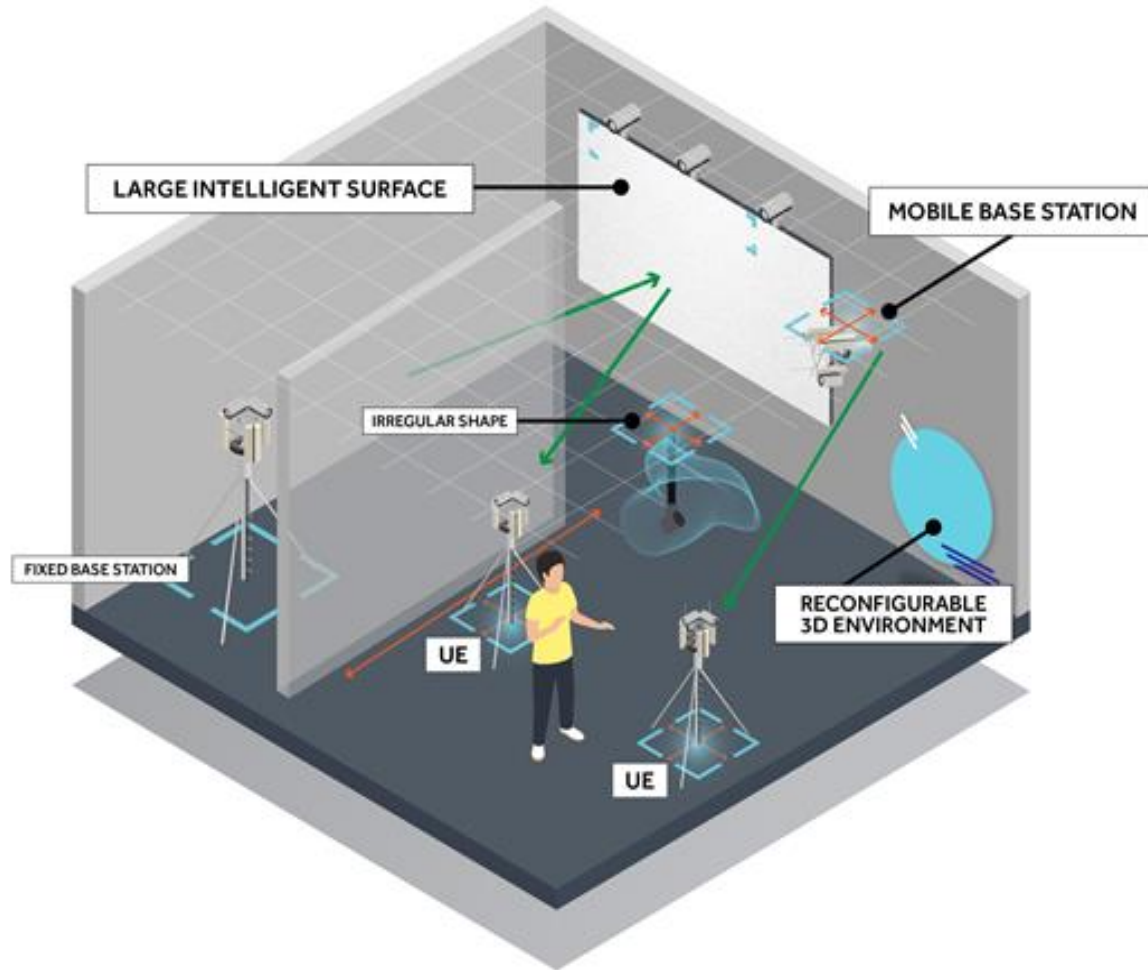
vision-radio 3D environment with radio propagation paths

4. Machine Learning algorithms for dataset analysis

- ▶ Set of **prediction algorithms** for several scenarios possibly faced by a user
- ▶ **Identifying objects** on images, **learning incrementally** from a data stream, or even **learning to interact with a dynamic environment**
- ▶ Implements base algorithms such as
 - **online learning** for classification and regression
 - **reinforcement learning**
 - **swarm intelligence** for dynamic scenarios and multiple-agents
 - **Deep Learning** for object detection
- ▶ **Main requirements**
 - ▶ Data anonymisation at the data source
 - ▶ Data quality, quantity, granularity
 - ▶ Computational resources



Proposed experimental chamber concept



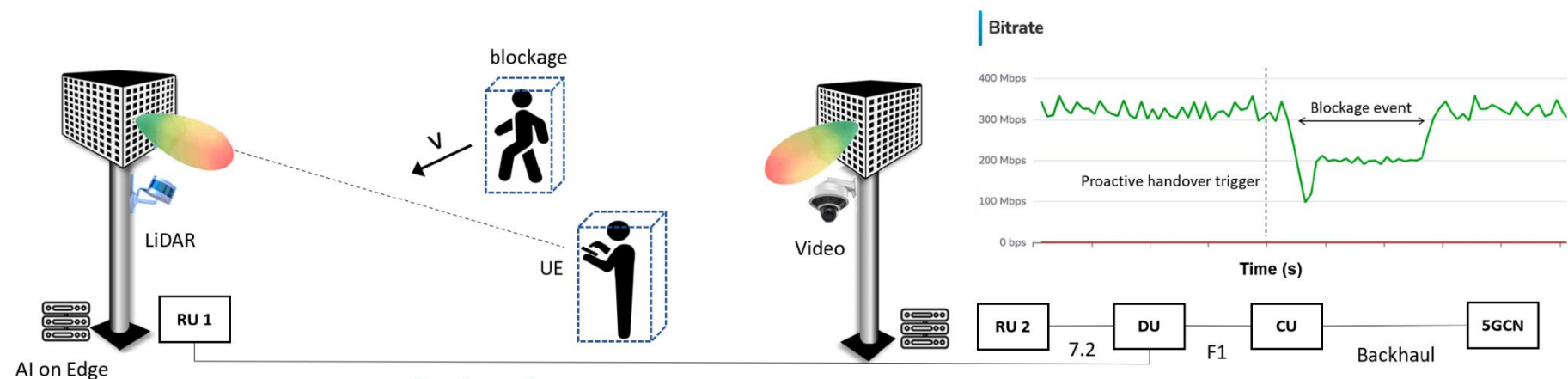
Mockup of the Graphical User Interface

- Marker based system for ground-truth acquisition
- RGB-D Cameras: Nerian Ruby, PTP

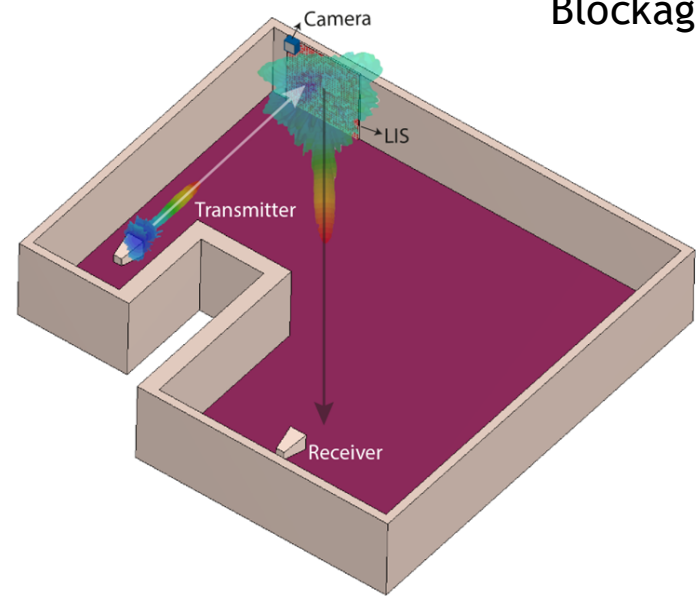
Main outcomes

- ▶ **Better understanding of joint radio, vision, sensing and traffic information**
- ▶ **ML models developed for jointly handling these types of data**
- ▶ **Better communications solutions** that take advantage of vision and sensing information
- ▶ **Better vision solutions** that take advantage of radio communications and sensing information
- ▶ **Enable new classes of applications** in the verticals of
 - Telecommunications
 - Automotive
 - Manufacturing
 - Media
 - Health

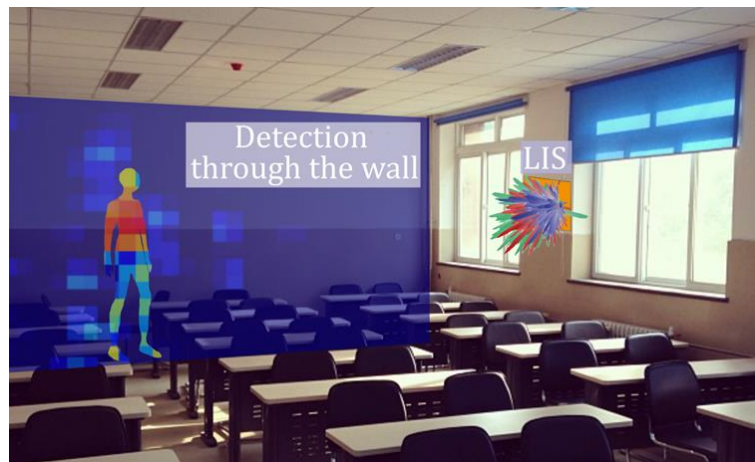
Use cases: Telecommunications



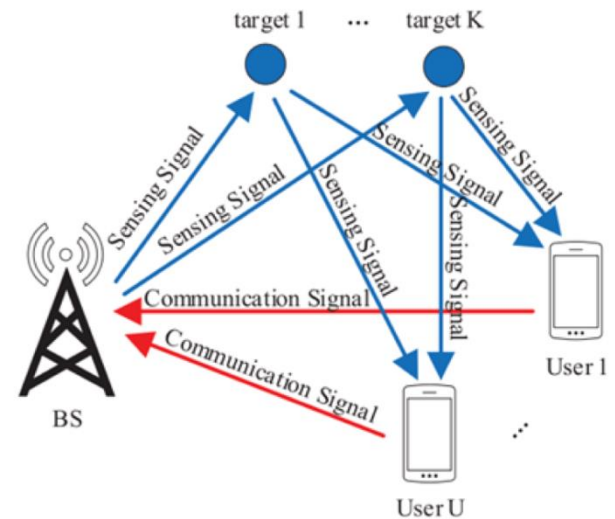
Blockage prediction and proactive handover



Coverage Enhancement



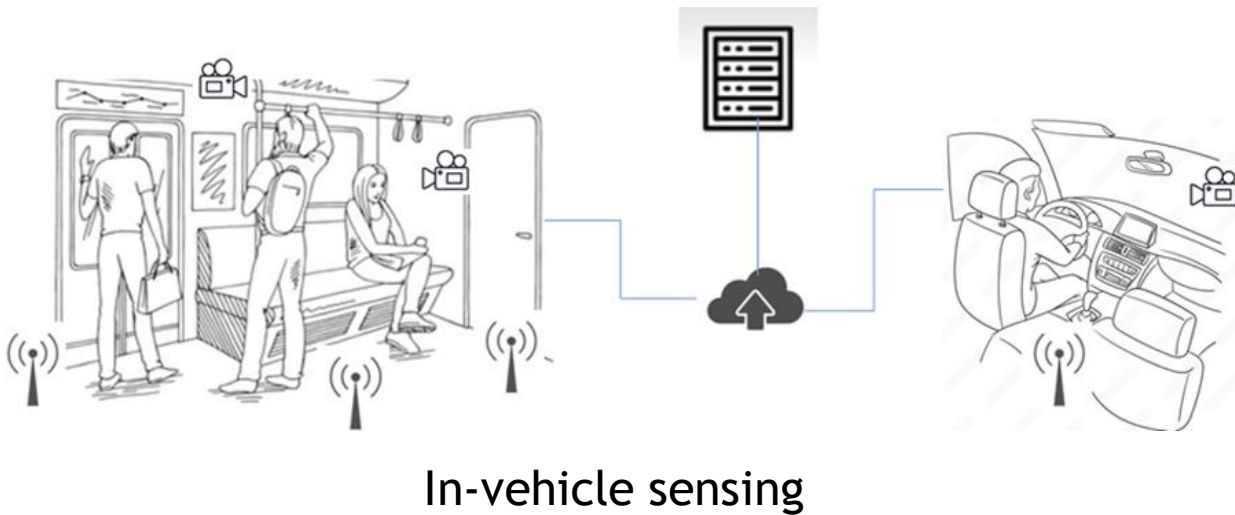
RF Mapping for Coverage Extension



Transparent Sensing for Enablement of Enhanced Radio Resource Management

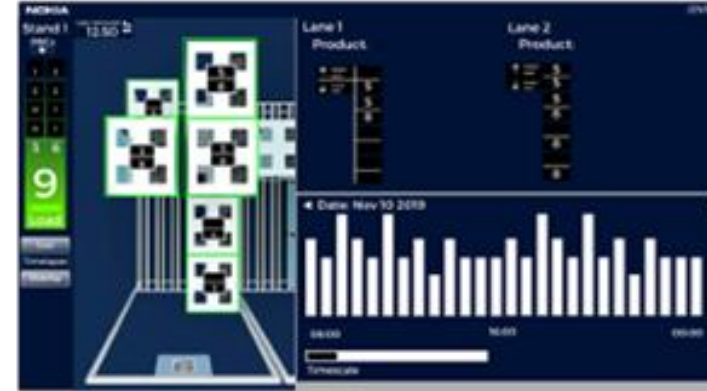
Use cases: Automotive

- Enhanced safety with sensor fusion
- V2V communication
- predictive hazard detection

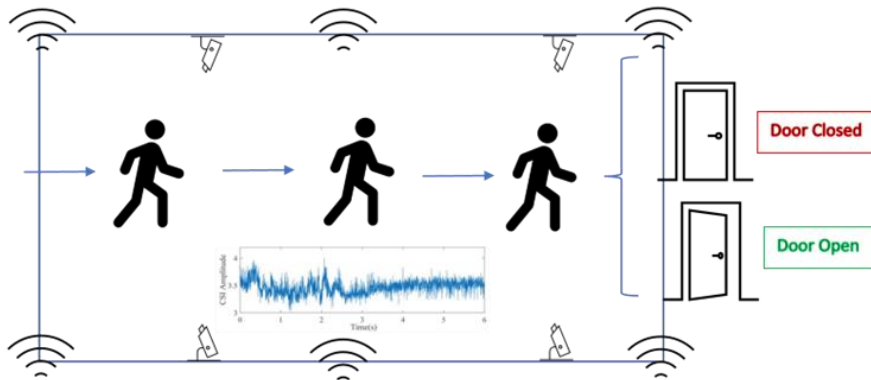


Use cases: Manufacturing

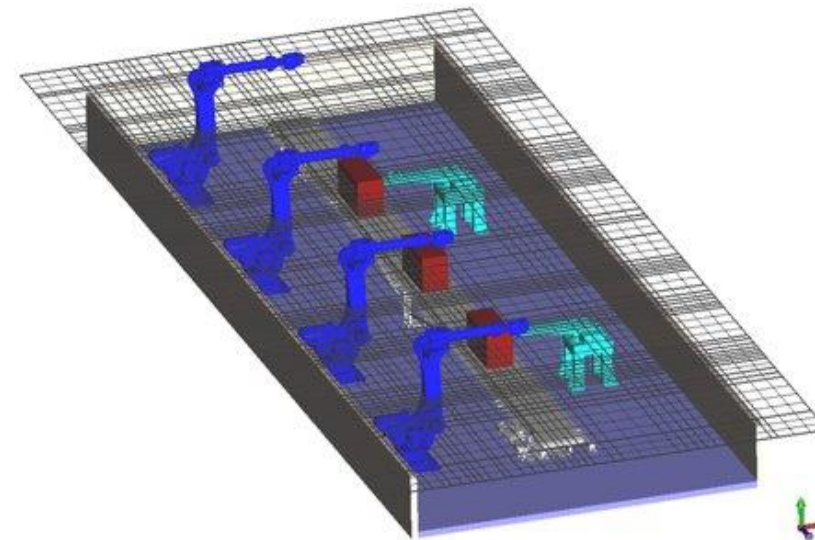
- Smart factory systems;
- Self-optimizing production lines;
- Integrated sensor networks.



Automated production lines with interconnected robots

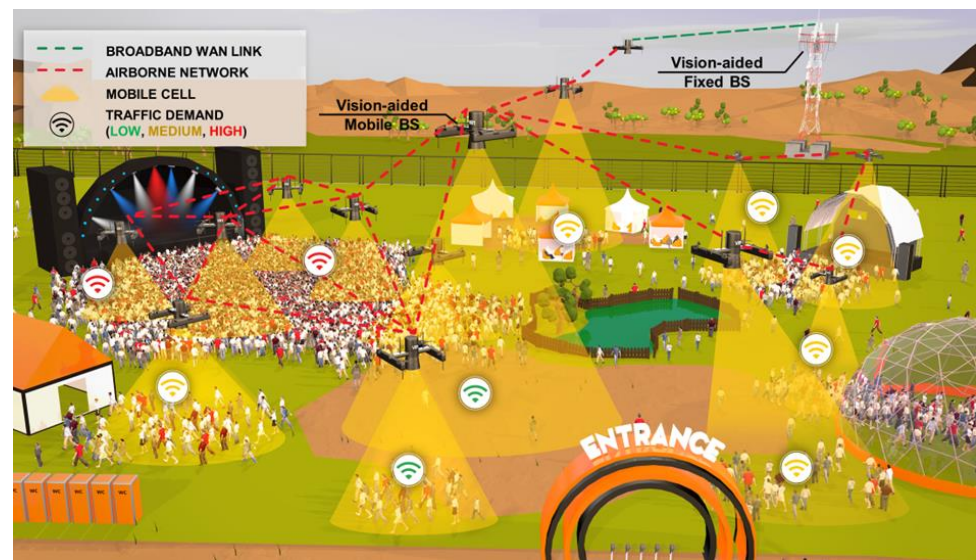


Privacy preserving human identification through RF sensing

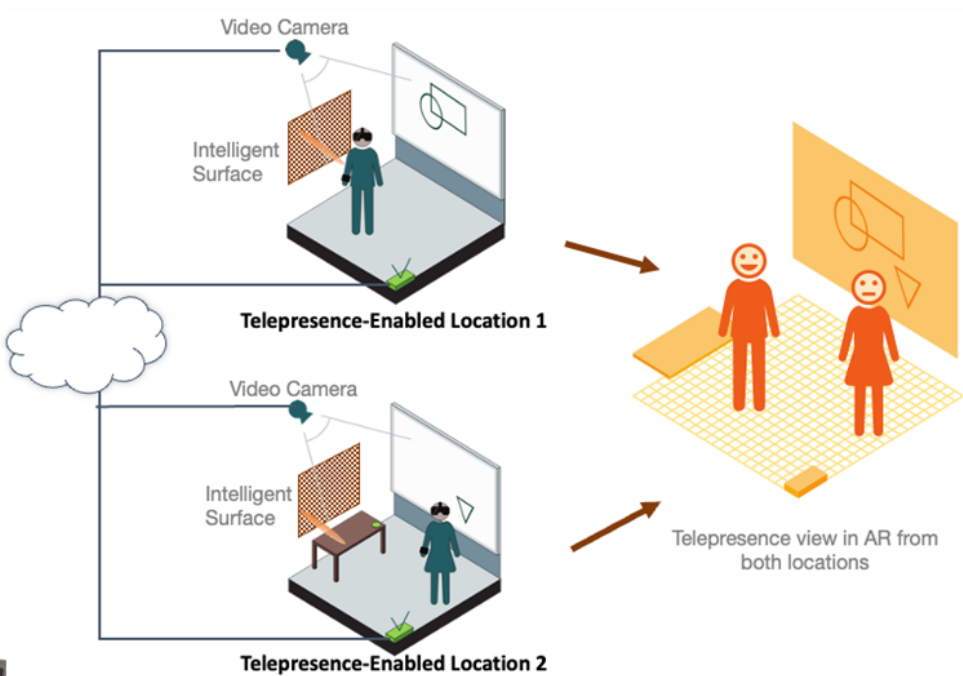


Digital Twinning

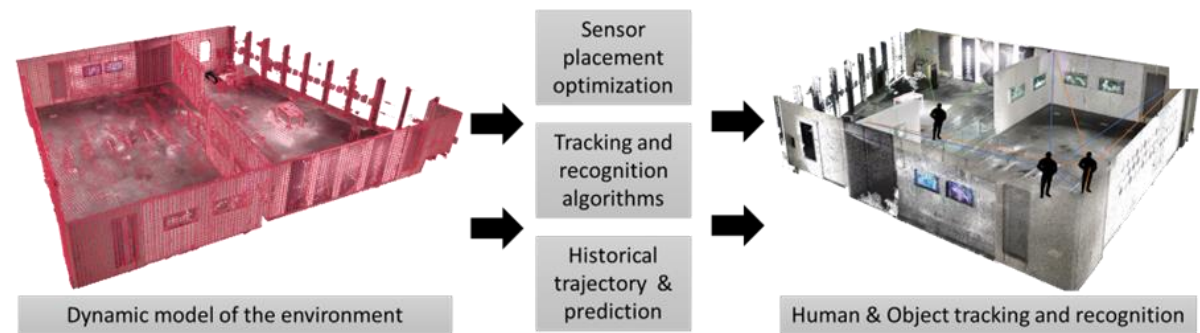
Use cases: Media



Outdoor vision-aided mobile base station



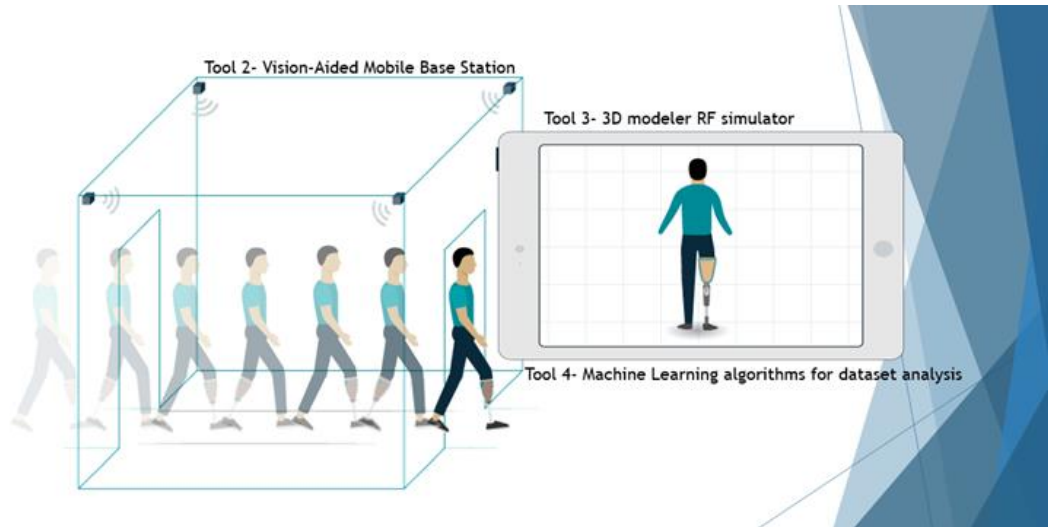
Immersive telepresence



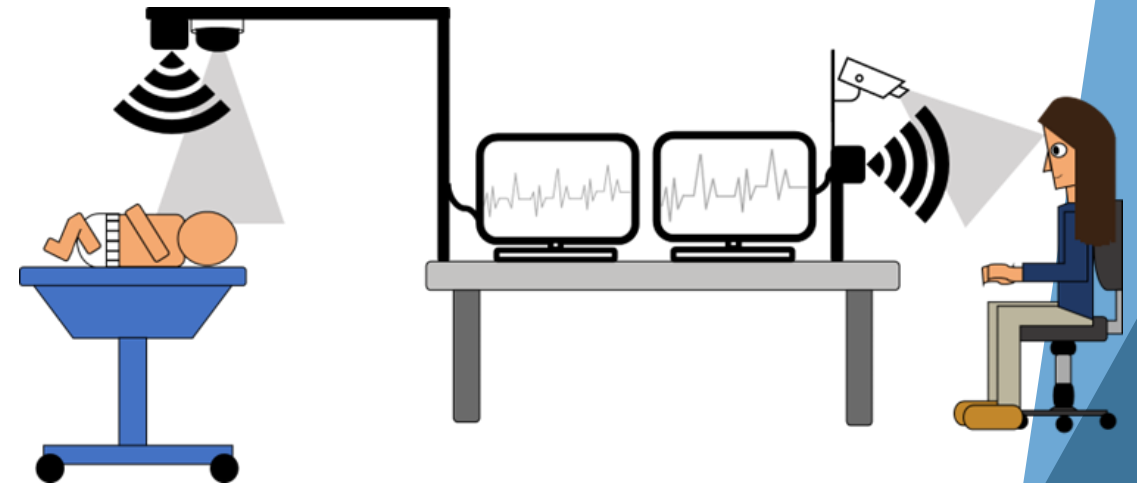
Motion tracking and object recognition (in smart home/office)

Use cases: Health

- Real-time, high-resolution imaging
- Telemedicine advancements
- Remote diagnostics support

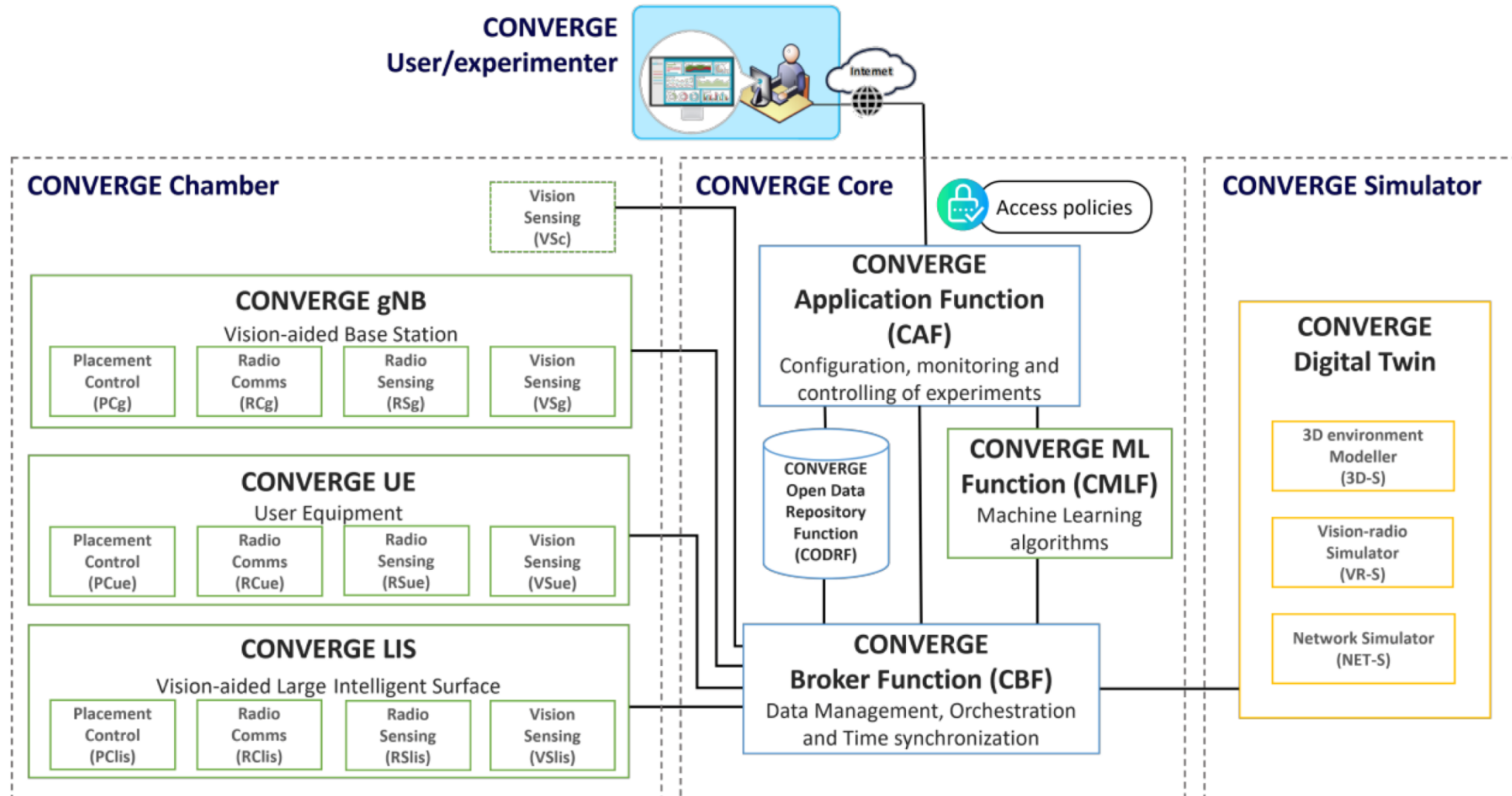


Vision+Radio patient monitoring



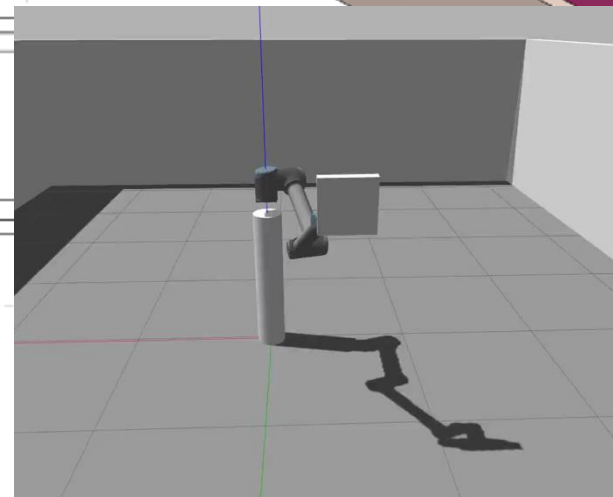
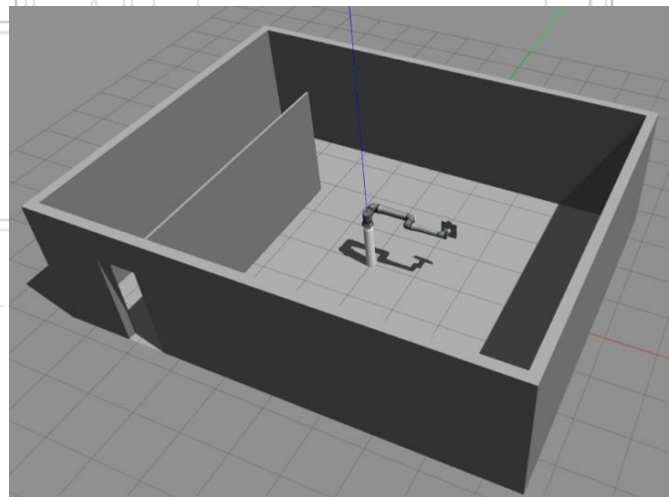
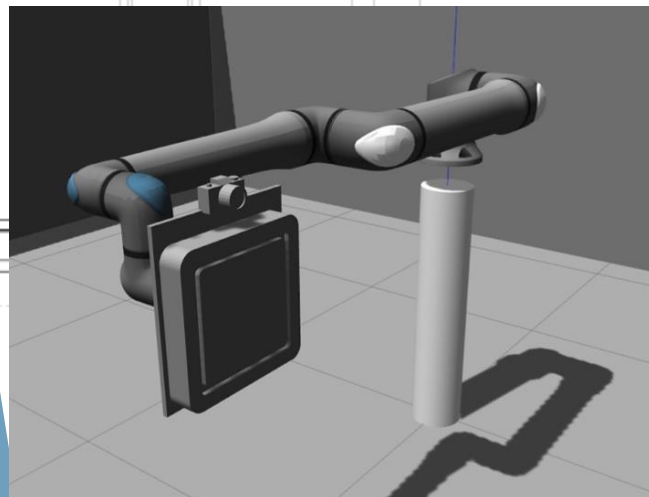
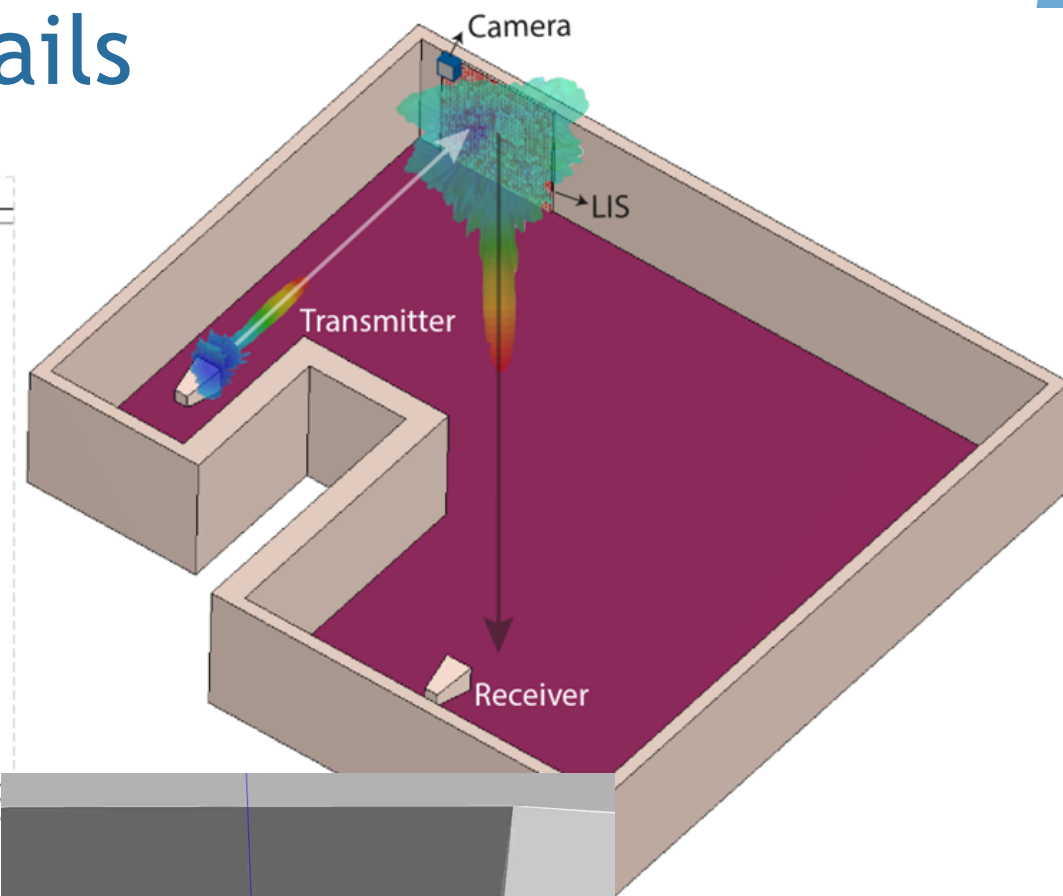
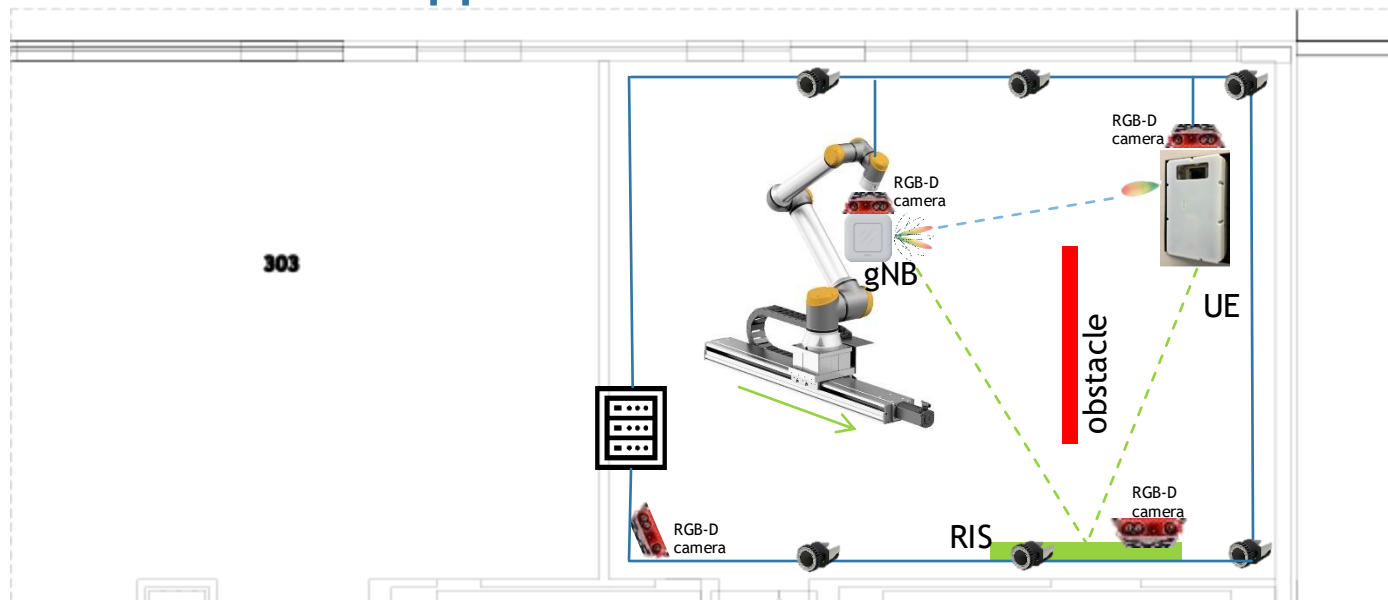
RF signals to extract breathing and heartbeats

CONVERGE high-level architecture



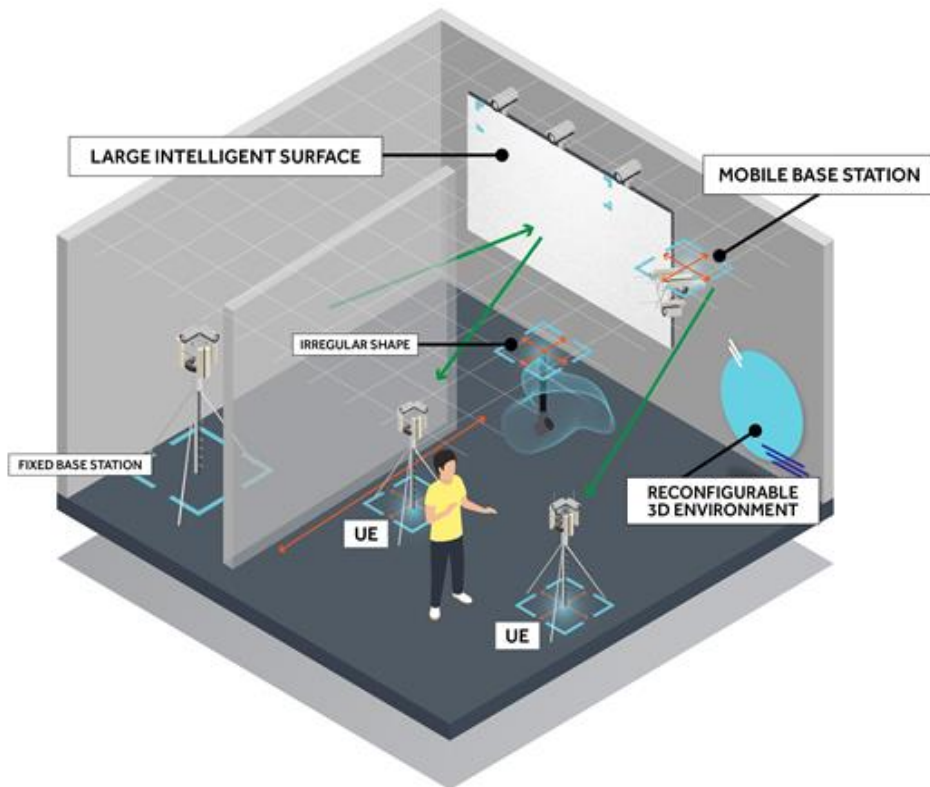
Chamber implementation details

► Possible approach at Porto site



CONVERGE

view-to-communicate and communicate-to-view



Thank you for your attention!